NGUYEN VIET KHANH

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EDUCATION

Université Paris-Saclay 2025 - Present
Université Évry Paris Saclay *Master 2 - Smart Aerospace and Autonomous System*

- **Scholarship**: IDEX scholarship

Hanoi University of Science and Technology *(HUST)* 2021 - 2025
School of Electrical Electronic Engineering *Bachelor - Control Engineering and Automation*

- **Cumulative GPA**: 3.72/4.0, **Ranking**: 43/478,
- **Award**: Best thesis defend award, Student Research Conference (HUST) 2nd Place (2025) and 3rd Place (2024)
- **Scholarship**: VATM Excellent student scholarship

EXPERIENCE

Member of the Motion Control and Applied Robotics Lab Jul 2022 - Aug 2025
Supervised by Assoc.Prof.Dr Tung Lam Nguyen, and Dr. Thi Van Anh Nguyen *Research Student*

- **Research direction**: Anti-sloshing for liquid sloshing cylindrical containers; Simulation for robotic systems
- Investigated **flatness-based trajectory** planning for anti-sloshing control.
- Developed control algorithms and validated models using **MATLAB/Simulink**.
- Programmed embedded systems in **C/C++** and performed physical simulation with **MuJoCo**.
- Built a solid background in control theory, motion control, and optimization.
- Experienced in scientific writing using **LaTeX** and presenting results at international conferences.

Member of the Computational Intelligence and Robotics Lab from Feb 2025 - Jun 2025
Supervised by Prof. Chen-Chien Hsu *Internship Student*

- **Research direction**: Multi-objective crowd-aware robot navigation using deep reinforcement learning
- Implemented learning-based algorithms and evolutionary computation techniques in **Python**.
- Simulated autonomous robots and tested navigation strategies in **ROS**.
- Gained adaptability and teamwork skills in an international research environment.

SKILLS AND RESEARCH INTEREST

Languages	English (IELTS 7.0)	**Simulation**	Simulink, Mujoco, ROS
Programming	Matlab, Python, C/C++	**Other tools:**	LaTeX, Git, Linux

Research interest: Nonlinear control, differential flatness, model predictive control, trajectory optimization, Multi-agent systems. Applications in motion control, robotic systems, UAV-UGV.

SELECTED PUBLICATIONS

1 **Khanh Nguyen Viet**, Hue Luu Thi, Thanh Cao Duc, Huy Nguyen Danh, Minh Nhat Vu, and Tung Lam Nguyen Time-optimal motion planning and anti-sloshing control for a container under disturbances, IEEE Access (**Q1 Scopus**, doi.org/10.1109/ACCESS.2025.3533541)

2 **Khanh Nguyen Viet**, Minh Do Duc, Thanh Cao Duc and Lam Nguyen Tung "Anti-sloshing control: Flatness-based trajectory planning and tracking control with an integrated extended state observer", IET Cyber-Systems and Robotics (**Q2 Scopus**, doi.org/10.1049/csy2.12121)

3 **Viet Khanh Nguyen**, Hue Luu Thi, Duc Thanh Cao, Dang Huu Bang, and Tung Lam Nguyen "The Non-Flatness Property of the Liquid Sloshing System and an Approximate Approach" 2024 International Conference on Advanced Technologies for Communications Program (doi.org/10.1109/ATC63255.2024.10908250)

4 **Khanh Nguyen Viet**, Hue Luu Thi, Minh Do Duc, Thanh Cao Duc, Huy Nguyen Danh, and Nguyen Tung Lam "Input Shaping Integrated With Lyapunov Based Model Predictive Control For Anti-Sloshing Problem" 2024 3rd *International Conference on Advances in Information and Communication Technology* (**Q4 Scopus**,doi.org/10.1007/978-3-031-80943-9_94)